



Assignment: Function Overloading in Java

Objective

To understand and implement **function overloading** by creating multiple methods with the same name but different parameter lists, based on real-world scenarios.

Instructions

- Method names should remain the same; only parameters should vary
- Use appropriate data types and method signatures
- Display clear output for each operation

Question 1: Calculator Using Function Overloading

Problem Statement

Create a class named **Calculator** that performs basic arithmetic operations.

Task

Implement the following operations using **function overloading**:

- Addition
- Subtraction
- Multiplication
- Division

Each operation must be overloaded to handle the following parameter combinations:

- Two `int` values
- Two `double` values
- One `int` and one `double`
- One `double` and one `int`

Question 2: Area Calculation Using Function Overloading

Problem Statement

Create a simple **area calculation system** using function overloading.



Classes

1. **Shape**
 - Attribute: (area)
2. **Triangle**
 - Attributes: (base, height)
3. **Rectangle**
 - Attributes: (length, breadth)
4. **Circle**
 - Attributes: (radius)

Requirement

- Create a method (e.g., calculateArea)
- Overload the method to calculate area for:
 - Triangle (base, height)
 - Rectangle (length, breadth)
 - Circle (radius)
 - Method overloading must be based on **parameter type** (Triangle, Rectangle, Circle)
 - Display the calculated area for each shape

Question 3: Loan Approval System Using Function Overloading

Problem Statement

Create a simple **loan approval system** using function overloading.

Classes

1. **Student**
 - Attributes: (rollNo, name, percentage)
2. **Employee**
 - Attributes: (id, name, annualSalary)
3. **Bank**
 - Contains overloaded methods to approve loans

Loan Approval Criteria

For Students (Based on Percentage)

- Above 80% → Loan Amount: ₹2,00,000
- 60% to 80% → Loan Amount: ₹1,00,000
- 40% to 60% → Loan Amount: ₹50,000
- Below 40% → No loan approved



For Employees (Based on Annual Salary)

- Above 12 LPA → Loan Amount: ₹7,00,000
- 10–12 LPA → Loan Amount: ₹6,00,000
- 6–10 LPA → Loan Amount: ₹5,00,000
- 4–6 LPA → Loan Amount: ₹4,00,000
- Below 4 LPA → No loan approved

Requirement

- Use the same method name (e.g., `approveLoan`)
- Overload methods based on **parameter type** (`Student` / `Employee`)

Question 4: Time Addition Using Function Overloading

Problem Statement

Create a class named **Time** to represent time using hours, minutes, and seconds.

Attributes (hr, min, sec)

Task

Overload the `add()` method to perform the following operations:

1. Add **two Time objects**
2. Add a **Time object and an integer**, where the integer can represent:
 - Hours **or**
 - Minutes **or**
 - Seconds

Output Requirement

- Display the result in the format:

hh:mm:ss

- Ensure proper time calculation (carry forward seconds and minutes correctly)
- Time must be displayed in **24-hour format**